

Industrial Asset Reliability & Maintenance Intelligence
Report
Process Unit B - Rotating Equipment Reliability Review
Synthetic Industrial Dataset for PIA Industrial Demonstration
Reporting Period: January–July 2026 | Report Date: 20 July 2026

Document Control

1. Executive Summary

This report consolidates operational observations, maintenance history, inspection findings, failure events, measurements, and recommended actions for selected assets in Process Unit B. The primary investigation concerns centrifugal process pump P-204, which transfers process fluid from storage tank T-204 through heat exchanger HX-204. Between May and July 2026, P-204 exhibited a progressive increase in vibration. A scheduled lubrication inspection was postponed on 30 June due to production requirements. On 16 July, operators detected abnormal mechanical noise, elevated drive-end vibration of 7.4 mm/s, and a bearing temperature of 91 degrees C. Inspection identified deterioration of the drive-end bearing, insufficient grease quantity, and degraded lubricant condition. The drive-end bearing was replaced on 17 July and lubrication was restored. Following maintenance, vibration decreased to 2.4 mm/s and bearing temperature stabilized at 66 degrees C. The available evidence supports a relationship between degraded lubrication condition and bearing deterioration. The postponed inspection represents a relevant maintenance-history event, but this report does not claim that postponement alone caused the failure. Additional observations are included for standby pump P-205 and heat exchanger HX-204 to provide a broader operational context. Recommended actions focus on lubrication inspection, vibration monitoring, maintenance deferral review, and evidence retention.

2. Facility and Process Context

Process Unit B contains storage, pumping, and heat-transfer equipment supporting continuous process operations. Storage tank T-204 provides process fluid to centrifugal pump P-204. P-204 supplies flow toward heat exchanger HX-204. Pump P-205 is designated as a standby unit for selected operating conditions. The reliability review covers equipment relationships only to the extent documented here. Asset identifiers should be treated as canonical references throughout this report.

Asset T-204 - Storage Tank. Function: holds process fluid upstream of P-204.
Asset P-204 - Centrifugal Process Pump. Function: transfers fluid from T-204 toward HX-204.
Asset P-205 - Standby Centrifugal Pump. Function: backup pumping capability.
Asset HX-204 - Shell-and-Tube Heat Exchanger. Function: process heat transfer downstream of the pumping stage.

3. Asset Register

P-204 is a centrifugal process pump driven by an electric motor. Its monitored indicators include drive-end vibration, non-drive-end vibration, bearing temperature, discharge pressure, and observable mechanical noise. P-205 is maintained as a standby pump. During the reporting period, no confirmed failure event was recorded for P-205. A routine inspection on 10 July noted acceptable external condition and no visible leakage. HX-204 is a heat exchanger associated with the downstream process path. A 5 July inspection recorded minor external fouling around an accessible inspection area. No loss-of-containment event was documented. T-204 is the upstream storage tank. No critical anomaly was documented during the reporting period covered by this report.

4. P-204 Operational Timeline

12 January 2026 - Routine visual inspection completed. No leakage or abnormal mechanical noise recorded.

14 February 2026 - Vibration reading recorded at 2.3 mm/s. Bearing temperature recorded at 64 degrees C. Condition assessed as normal.

12 March 2026 - Routine inspection completed. No abnormal vibration detected. Lubrication condition recorded as acceptable.

18 April 2026 - Vibration reading 2.6 mm/s. Bearing temperature 65 degrees C. No corrective work initiated.

28 May 2026 - Vibration reading increased to 3.9 mm/s. The observation was documented as a minor increase relative to earlier readings. Continued monitoring was recommended.

15 June 2026 - Vibration recorded at 4.6 mm/s. Bearing temperature recorded at 72 degrees C. Maintenance review requested.

30 June 2026 - Planned lubrication inspection postponed due to production requirements. No replacement inspection date was recorded in the maintenance note.

8 July 2026 - Operator shift log noted intermittent mechanical noise near P-204 during steady operation.

14 July 2026 - Vibration measured at 6.2 mm/s. Bearing temperature measured at 82 degrees C. Maintenance notification raised.

16 July 2026, 14:35 - Abnormal mechanical noise reported. Drive-end vibration measured at 7.4 mm/s and bearing temperature reached 91 degrees C. Pump remained operating while maintenance personnel assessed the condition.

16 July 2026, 16:10 - Inspection found insufficient grease quantity and evidence of lubricant degradation at the drive-end bearing location. Bearing deterioration was suspected.

17 July 2026 - Drive-end bearing replaced. Lubrication restored. Alignment and basic post-maintenance checks completed.

18 July 2026 - Post-maintenance vibration measured at 2.4 mm/s. Bearing temperature stabilized at 66 degrees C. No abnormal mechanical noise recorded during the observation period.

5. Measurement Record

The following measurements are observations recorded for demonstration and analysis.

14 February 2026: P-204 vibration 2.3 mm/s; bearing temperature 64 degrees C.

18 April 2026: P-204 vibration 2.6 mm/s; bearing temperature 65 degrees C.

28 May 2026: P-204 vibration 3.9 mm/s; bearing temperature 68 degrees C.

15 June 2026: P-204 vibration 4.6 mm/s; bearing temperature 72 degrees C.

14 July 2026: P-204 vibration 6.2 mm/s; bearing temperature 82 degrees C.

16 July 2026: P-204 vibration 7.4 mm/s; bearing temperature 91 degrees C.

18 July 2026: P-204 vibration 2.4 mm/s; bearing temperature 66 degrees C.

The measurements show an increasing vibration and temperature trend before maintenance, followed by lower

values after bearing replacement and lubrication restoration. These observations provide temporal evidence;

interpretation of causality should remain tied to documented inspection findings and validated analytical logic.

6. Failure Investigation - P-204

Event classification: Bearing degradation / abnormal vibration.

Affected asset: P-204.

Affected component: Drive-end bearing.

Detection method: Operator observation and condition measurements.

Observed symptoms: Abnormal mechanical noise, elevated vibration, elevated bearing temperature.

Inspection findings: Insufficient grease quantity, degraded lubricant condition, deteriorated drive-end bearing.

Immediate action: Bearing replacement and lubrication restoration.

Post-action condition: Vibration and temperature returned near previously observed operating levels.

The evidence supports inadequate lubrication as a plausible contributor to bearing degradation. The maintenance

history also records a postponed lubrication inspection. Because multiple factors can contribute to bearing

degradation, the postponement should be treated as contextual evidence rather than an independently proven root

cause unless supported by additional analysis.

7. Maintenance Work Order Summary

Work Order WO-P204-0717

Asset: P-204

Priority: High

Opened: 16 July 2026

Completed: 17 July 2026

Scope: Inspect drive-end bearing, lubrication condition, coupling area, and vibration condition.

Finding: Drive-end bearing showed deterioration. Lubrication quantity was insufficient and lubricant condition was degraded.

Action: Replaced drive-end bearing. Restored lubrication. Performed operational check.

Verification: Vibration 2.4 mm/s and bearing temperature 66 degrees C on 18 July.

Status: Completed.

Deferred Task MT-P204-0630

Asset: P-204

Original task: Lubrication inspection.

Scheduled date: 30 June 2026.

Disposition: Postponed due to production requirements.

Follow-up date in original note: Not recorded.

Status at time of failure event: Outstanding.

8. Standby Pump P-205

P-205 was inspected on 10 July 2026 as part of standby equipment readiness checks. External condition was acceptable. No visible leakage was reported. Manual rotation and accessible inspection checks did not identify a confirmed defect.

A readiness test is recommended following significant maintenance activity on the primary pump P-204 so that standby capability is supported by current evidence. This recommendation is preventive and does not indicate a known failure of P-205.

9. Heat Exchanger HX-204 Inspection

On 5 July 2026, maintenance personnel recorded minor external fouling around an accessible section of HX-204. No confirmed process leak or loss of containment was documented. The observation was classified for routine follow-up.

Recommended action: inspect the affected external area during the next planned maintenance window and document whether fouling is stable, increasing, or associated with another observable condition.

10. Maintenance Pattern Review

The P-204 timeline contains several events relevant to reliability review: a minor vibration increase in May, a further increase in June, postponement of a lubrication inspection, abnormal noise in July, high vibration and temperature on

16 July, and recovery following bearing replacement and lubrication restoration.

This sequence is suitable for evidence-based pattern analysis because it contains timestamps, measurements, maintenance actions, component findings, and post-maintenance verification. Any automated system analyzing the sequence should preserve source provenance and distinguish observations from inferred conclusions.

11. Risk and Criticality Notes

P-204 is operationally important because it supports process-fluid transfer between T-204 and HX-204. Loss of

pumping capability could interrupt the associated process path. The presence of standby pump P-205 may provide

operational resilience, subject to confirmed readiness.

For this synthetic report, P-204 should be treated as a critical rotating asset for maintenance prioritization. This

designation is provided as report context and should not be interpreted as a formal plant-wide quantitative criticality assessment.

12. Compliance and Record-Retention Review

Maintenance and inspection records for critical rotating equipment should retain the asset identifier, event timestamp, observed condition, measurement values where available, maintenance action, responsible work order, and post-maintenance verification.

The 30 June deferred lubrication task did not contain a documented replacement inspection date in the source note.

The maintenance process should require deferred critical tasks to include a revised due date and accountable follow-up.

This report does not assert compliance with a specific external regulation or standard. It provides internal maintenance-control observations suitable for demonstration of evidence traceability and compliance-oriented review.

13. Recommendations

R1 - Establish a recurring lubrication-condition inspection for P-204 at an interval approved by the facility maintenance program.

R2 - Continue vibration and bearing-temperature trending for P-204. Investigate material deviation from the established normal operating range.

R3 - Review the process for postponing maintenance on critical rotating equipment. Require a documented revised due date and risk review for deferred tasks.

R4 - Perform a follow-up inspection of the replacement drive-end bearing within 60 days or according to the approved maintenance strategy.

R5 - Confirm standby readiness of P-205 after major corrective maintenance on P-204.

R6 - Retain measurement and inspection provenance so that future conclusions can be traced to source events.

R7 - Review HX-204 external fouling during the next planned maintenance opportunity and document the result.

14. Evidence Register

EV-001: 28 May vibration observation - P-204 - 3.9 mm/s - source type: condition reading.

EV-002: 15 June vibration observation - P-204 - 4.6 mm/s - source type: condition reading.

EV-003: 30 June lubrication inspection deferral - P-204 - source type: maintenance record.

EV-004: 8 July intermittent noise observation - P-204 - source type: shift log.

EV-005: 14 July elevated vibration and temperature - P-204 - source type: condition reading.

EV-006: 16 July abnormal noise and high vibration - P-204 - source type: operator event.

EV-007: 16 July insufficient grease and degraded lubricant - P-204 drive-end bearing - source type: inspection finding.

EV-008: 17 July bearing replacement and lubrication restoration - P-204 - source type: work order.

EV-009: 18 July vibration 2.4 mm/s and temperature 66 degrees C - P-204 - source type: post-maintenance verification.

EV-010: 10 July standby inspection - P-205 - source type: inspection record.

EV-011: 5 July external fouling observation - HX-204 - source type: inspection record.

15. Suggested Questions for an Industrial Intelligence Platform

What assets are identified in this report?

What happened to pump P-204?

Show the timeline of P-204 observations and maintenance.

Which measurements changed before the failure event?

What evidence is associated with the drive-end bearing?

Was a maintenance activity postponed before the event?

What actions were performed after the high-vibration event?

What changed after maintenance?
What recommendations remain open?
Which evidence supports the lubrication-related finding?
What is known about standby pump P-205?
Are there compliance-oriented documentation gaps?
What relationships exist among T-204, P-204, and HX-204?

16. Data Quality and Interpretation Notes

All asset identifiers, dates, measurements, and events in this document are synthetic and created for software demonstration. The report intentionally contains a mixture of direct observations, measurements, maintenance actions, recommendations, and cautious interpretations. Systems processing this report should avoid converting recommendations into completed actions, avoid treating contextual events as proven causes, and preserve distinctions between observed facts and analytical conclusions. Where evidence is insufficient, the correct result is an explicit insufficient-evidence state rather than a fabricated conclusion.

17. Conclusion

The P-204 case demonstrates a realistic industrial reliability sequence: normal operation, progressive condition change, a deferred maintenance activity, a significant abnormal event, component inspection, corrective maintenance, and post-maintenance verification. The dataset is designed to test whether an industrial intelligence platform can transform fragmented operational information into structured, traceable knowledge while maintaining evidence provenance. It also provides related assets and secondary observations to support graph construction and workspace-level analysis without relying on proprietary plant information.

Appendix A - Synthetic Shift Log Extracts

LOG-001: 08 July 2026, Day Shift: P-204 operating. Intermittent mechanical noise noted near drive-end area. No visible leakage. Maintenance notification recommended.
LOG-002: 14 July 2026, Evening Shift: Increased vibration reported on P-204. Reading recorded at 6.2 mm/s. Bearing temperature 82 degrees C. Maintenance team informed.
LOG-003: 16 July 2026, Day Shift: Abnormal mechanical noise became persistent. Vibration measured at 7.4 mm/s. Bearing temperature reached 91 degrees C. Maintenance assessment initiated.
LOG-004: 18 July 2026, Day Shift: P-204 returned to service following bearing replacement. Vibration 2.4 mm/s. Bearing temperature 66 degrees C. No abnormal noise observed.

Appendix B - Demonstration Disclaimer

This document is entirely synthetic. It does not describe a real facility, company, incident, customer, or regulated asset. It is intended solely as a realistic test document for PIA Industrial ingestion, evidence extraction,

knowledge-graph construction, traceability, maintenance intelligence, failure analysis, compliance-oriented review, and decision-support demonstrations.